

Circular Convolution:

What results from multiplying two N -point DFTs?

$$\begin{aligned}
 X[k]W[k] &= \sum_{m=0}^{N-1} x[m] e^{-j\frac{2\pi}{N}mk} \sum_{l=0}^{N-1} w[l] e^{-j\frac{2\pi}{N}lk} \\
 &= \sum_{m=0}^{N-1} x[m] \sum_{l=0}^{N-1} w[l] e^{-j\frac{2\pi}{N}(m+l)k} \quad \leftarrow m \text{ is fixed.}
 \end{aligned}$$

Using the substitution

$$\langle n-m \rangle_N = n-m - \left\lfloor \frac{n-m}{N} \right\rfloor N$$

$$l = \langle n-m \rangle_N = n-m + qN \text{ for some } q \in \mathbb{Z}$$

we get

$$q = - \left\lfloor \frac{n-m}{N} \right\rfloor$$

$$e^{-j2\pi qk} = 1$$

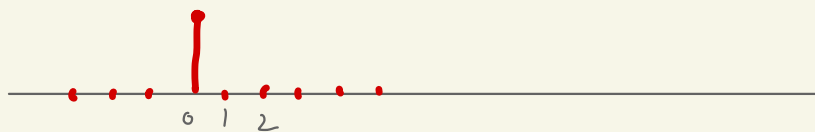
$$\begin{aligned}
 X[k]W[k] &= \sum_{m=0}^{N-1} x[m] \sum_{n=0}^{N-1} w[\langle n-m \rangle_N] \underbrace{e^{-j\frac{2\pi}{N}(m+n-m+qN)k}}_{e^{-j\frac{2\pi}{N}nk}} \\
 &= \sum_{n=0}^{N-1} \left(\sum_{m=0}^{N-1} x[m] w[\langle n-m \rangle_N] \right) e^{-j\frac{2\pi}{N}nk} \\
 &\stackrel{N\text{-DFT}}{\longleftrightarrow} \sum_{m=0}^{N-1} x[m] w[\langle n-m \rangle_N] \triangleq \{x[n]\} \circledast \{w[n]\}
 \end{aligned}$$

DFT of $\square$ at freq. k

known as “ N -circular convolution.”

convolution of $x[n]$ and $w[\langle n \rangle_N]$

$f[n]$



$f[\langle n \rangle_N]$



$x[n] \rightarrow x[\langle n \rangle_N]$ is periodization of $x[n]$
supported within
 $\{0, 1, \dots, N-1\}$

OR

$x[n] = 0$ if $n < 0$ OR $n \geq N$

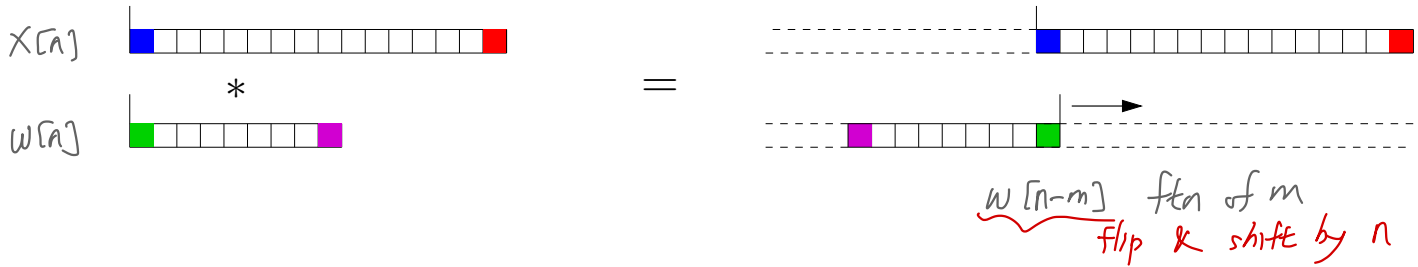
$$x[\langle n + N \rangle_N] = x[\langle n \rangle_N]$$

Circular Convolution (cont.):

Linear convolution:

$$y[n] = \sum_{m=-\infty}^{\infty} x[m]w[n-m]:$$

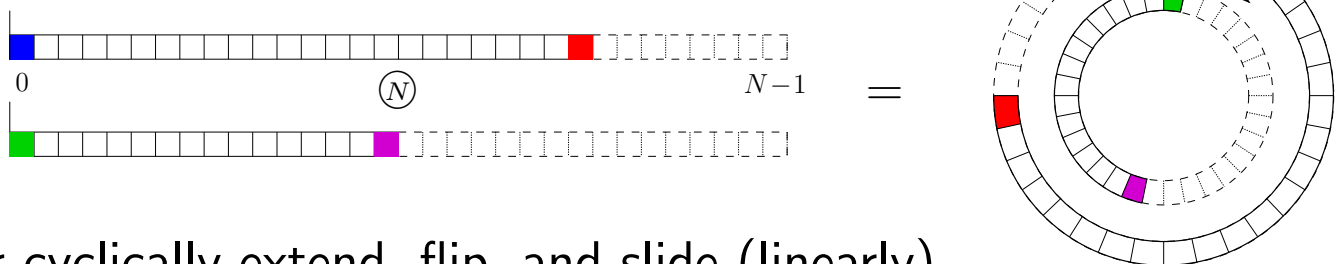
flip and slide (linearly)...



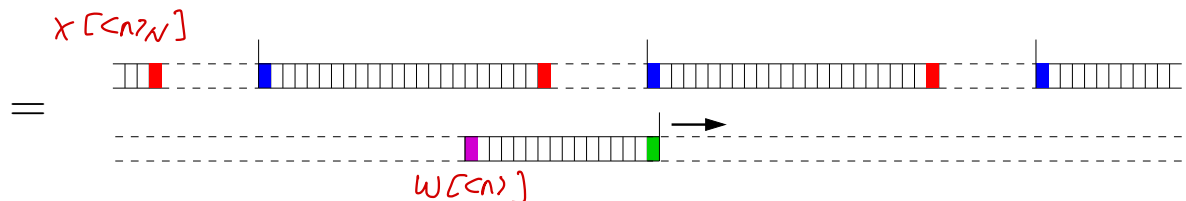
N -circular convolution:

$$y[n] = \sum_{m=0}^{N-1} x[m]w[\langle n-m \rangle_N]:$$

flip and slide (cyclically)...



or cyclically extend, flip, and slide (linearly)...



$$y[n] = \sum_{m=-\infty}^{\infty} x[\langle m \rangle_N]w[n-m]$$

(Assume finite-length $\{w[m]\}$ is zero-extended in last equation.)

Circular Convolution — Examples:

1. Consider $[1, 2, 3] \circledast_N [2, -1, 1]$ for $N = 3$:

1	2	3	1	2	3	1	2	3	out:
	1	-1	2						1
		1	-1	2					6
			1	-1	2				5
				1	-1	2			1
					1	-1	2		6
						1	-1	2	5
							$\ddots$		$\vdots$

2. Now consider $[1, 2, 3] \circledast_N [2, -1, 1]$ for $N = 5$,
or, equivalently, $[1, 2, 3, 0, 0] \circledast_N [2, -1, 1, 0, 0]$ for $N = 5$:

1	2	3	0	0	1	2	3	0	0	out:
	0	0	1	-1	2					2
		0	0	1	-1	2				3
			0	0	1	-1	2			5
				0	0	1	-1	2		-1
					0	0	1	-1	2	3
						$\ddots$				$\vdots$

$$x[n] \quad \{1, 2, 3\}$$

$$n = 0 \quad 1 \quad 2$$

$$w[n] \quad \{2, -1, 1\}$$

$$n = 0 \quad 1 \quad 2$$

$$w[n] * x[n]$$

$$\begin{bmatrix} 2 & 0 & 0 \\ -1 & 2 & 0 \\ 1 & -1 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Toeplitz

circulant matrix

$$N = 5$$

$$\begin{array}{cccccc} x & 1 & 2 & 3 & 0 & 0 \\ w & 2 & -1 & 1 & 0 & 0 \end{array}$$

$$\begin{bmatrix} 2 & 0 & 0 & 1 & -1 \\ -1 & 2 & 0 & 0 & 1 \\ 1 & -1 & 2 & 0 & 0 \\ 0 & 1 & -1 & 2 & 0 \\ 0 & 0 & 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 0 \\ 0 \end{bmatrix}$$

Let $W \in \mathbb{C}^{N \times N}$ be the DFT matrix of size N .

$$\hat{x} = Wx$$

$$\hat{h} = Wh$$

$$x \otimes_N h = C_h x$$

↑
Circulant
matrix by h

$$= \underbrace{\begin{bmatrix} a_1 & a_2 & \dots & a_N \end{bmatrix}}_{\text{diag}(a)} \underbrace{\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_N \end{bmatrix}}_b$$

$$\underbrace{\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_N \end{bmatrix}}_a \odot \underbrace{\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_N \end{bmatrix}}_b = \underbrace{\begin{bmatrix} a_1 b_1 \\ a_2 b_2 \\ \vdots \\ a_N b_N \end{bmatrix}}_{a \odot b}$$

Hadamard product.

$$W(x \otimes_N h) = Wx \odot Wh = \text{diag}(Wh) Wx$$

$$C_h x = W^{-1} \text{diag}(Wh) Wx$$

$$C_h = W^{-1} \text{diag}(Wh) W$$

$$= \frac{1}{N} W^H \text{diag}(Wh) W$$

$$= \underbrace{\left(\frac{1}{\sqrt{N}} W \right)^H}_U \text{diag}(Wh) \underbrace{\left(\frac{1}{\sqrt{N}} W \right)}_{U^H}$$

$$U = \frac{1}{\sqrt{N}} W^H$$

$$U^H U = U U^H = I_N$$

$$x[n] = 0 \quad \text{if } n < 0 \text{ or } n \geq N$$

$$x[n] \xleftrightarrow{1-1} x[\langle n \rangle_N]$$

The DFT — important properties:

Here, $x[n] \xleftrightarrow{N\text{-DFT}} X[k]$, $w[n] \xleftrightarrow{N\text{-DFT}} W[k]$, and $n, k = 0 \dots N-1$.

$$X[k] \xleftrightarrow{1-1} X[\langle k \rangle_N]$$

1. Linearity:

$$\alpha x[n] + \beta w[n] \xleftrightarrow{N\text{-DFT}} \alpha X[k] + \beta W[k]$$

2. Conjugate symmetry:

$$x^*[n] = x[\langle -n \rangle_N] \Leftrightarrow X[k] \in \mathbb{R}$$

$$x[n] \in \mathbb{R} \Leftrightarrow X^*[k] = X[\langle -k \rangle_N]$$

3. Shift/modulation:

$$x[\langle n - d \rangle_N] \xleftrightarrow{N\text{-DFT}} e^{-j\frac{2\pi}{N}dk} X[k], \quad d \in \mathbb{Z}$$

$$e^{j\frac{2\pi}{N}dn} x[n] \xleftrightarrow{N\text{-DFT}} X[\langle k - d \rangle_N], \quad d \in \mathbb{Z}$$

4. Reversal:

$$x[\langle -n \rangle_N] \xleftrightarrow{N\text{-DFT}} X[\langle -k \rangle_N]$$

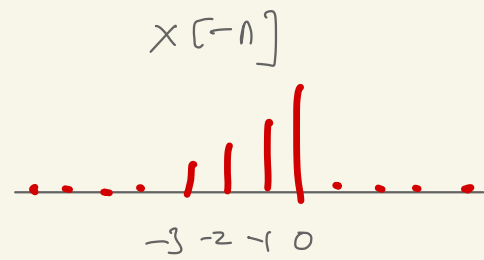
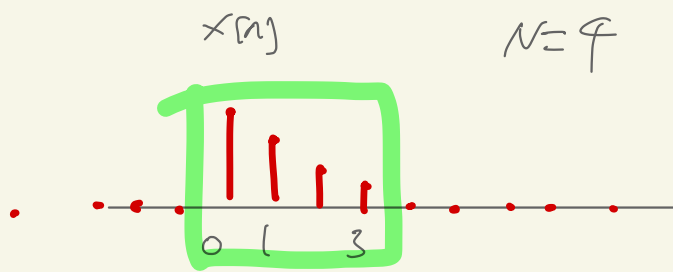
5. Multiplication/convolution:

$$x[n]w[n] \xleftrightarrow{N\text{-DFT}} \frac{1}{N} \sum_{l=0}^{N-1} X[l]W[\langle k - l \rangle_N]$$

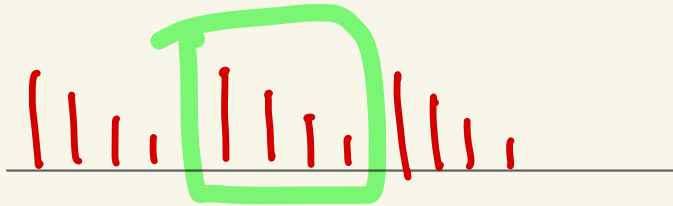
$$\sum_{m=0}^{N-1} x[m]w[\langle n - m \rangle_N] \xleftrightarrow{N\text{-DFT}} X[k]W[k]$$

6. Energy conservation (“Parseval’s theorem”):

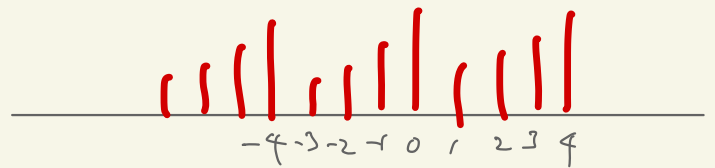
$$\sum_{n=0}^{N-1} |x[n]|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X[k]|^2$$



$$x[\langle n \rangle_N] = y[n]$$



$$x[\langle -n \rangle_N] = y[-n]$$



$$X(k) = \sum_{n=0}^{N-1} x[n] e^{-j \frac{2\pi}{N} kn}$$

$$= \sum_{n=0}^{N-1} x[\langle n \rangle_N] e^{-j \frac{2\pi}{N} kn}$$

$$\langle m \rangle_N = m - \left\lfloor \frac{m}{N} \right\rfloor N$$

modulo - N

$$m=7 \quad N=3$$

$$\left\lfloor \frac{7}{3} \right\rfloor = \left\lfloor 2.\overline{3} \right\rfloor = 2$$

Working with modulo indices:

- Many DFT properties involve modulo- N indices!
- Main idea:** $\langle m \rangle_N = m + qN$ for integer q (in fact $q = -\left\lfloor \frac{m}{N} \right\rfloor$, but the exact value will not matter). Recall circular convolution example (p. 7). Here's another...
- Proof of $x[\langle -n \rangle_N] \xleftrightarrow{N\text{-DFT}} X[\langle -k \rangle_N]$:

$$\begin{aligned}
 x[\langle -n \rangle_N] &\xleftrightarrow{N\text{-DFT}} \sum_{n=0}^{N-1} x[\langle -n \rangle_N] e^{-j\frac{2\pi}{N}nk} \\
 &\quad \text{for integer } q \\
 &\quad \underbrace{\langle -n \rangle_N}_{-n + qN \triangleq m} \\
 &= \sum_{m=0}^{N-1} x[m] e^{-j\frac{2\pi}{N}(qN-m)k} \quad \text{same sum!} \\
 &= \sum_{m=0}^{N-1} x[m] e^{-j\frac{2\pi}{N}m(-k)} \quad \leftarrow \text{DFT of } x[n] \text{ at } -k
 \end{aligned}$$

$m = \langle -n \rangle_N$

$\langle -n \rangle_N = -n - \left\lfloor \frac{-n}{N} \right\rfloor N$

Can't write " $X[-k]$ " because DFT is only defined for $k = 0 \dots N-1$! Instead, $\langle -k \rangle_N = -k + rN$ for $r \in \mathbb{Z}$:

$$\begin{aligned}
 -k \mapsto \langle -k \rangle_N &= \sum_{m=0}^{N-1} x[m] e^{-j\frac{2\pi}{N}m(\langle -k \rangle_N - rN)} \quad \text{for integer } r \\
 &= X[\langle -k \rangle_N]
 \end{aligned}$$

- Example:

$n :$	-5	-4	-3	-2	-1	0	1	2	3	4	5
$x[n] :$						a	b	c	d	e	
$x[-n] :$		e	d	c	b	a					
$x[\langle -n \rangle_5] :$	a	e	d	c	b	a	e	d	c	b	a

$$m = -7 \quad N = 3$$

$$\left\lfloor \frac{m}{N} \right\rfloor = \left\lfloor -2.\text{xx}\dots \right\rfloor = -3$$

$$-7 - (-3) \cdot 3 = 2$$

$$\begin{array}{ccccccc} \uparrow & & \uparrow & \uparrow & \uparrow & & \\ m & & \left\lfloor \frac{m}{N} \right\rfloor & N & \langle m \rangle_N & & \end{array}$$